

Machine Learning - Project

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1 Introduction

2 Pareto optimal states and Nash equilibria of matrix games

For each of the 4 games a matrix is given which describes players, their possible actions and utility functions. This is equivalent to a normal-form game tuple (N, A, u) as in [YS09]. N is the set of integers where each element represents a player. A is the set which represents the cartesian product of each player's possible actions and u is the utility function which takes an input an action profile (element of A) and returns a real number.

2.1 Important definitions

Definition 2.1. A strategy s of a player i is the probability distribution over its actions and S_i is the set of all strategies:

$$s \in S_i$$

Definition 2.2. Set of all strategy profiles S is a cartesian product of sets of strategies of every player:

$$S = S_1 \times \dots \times S_N$$

Definition 2.3. Expected utility for a player i given a strategy profile s :

$$e_i(s) = \sum_{a \in A} u_i(a) \prod_{j=1}^n s_j(a_j), \quad s \in S$$

Definition 2.4. Strategy profile s dominates strategy profile s' if:

a)

$$\forall i \in N : e_i(s) \geq e_i(s')$$

b)

$$\exists j \in N : e_j(s) > e_j(s')$$

If a strategy profile is not Pareto dominated than it is Pareto optimal.

2.2 Game 1 - Stag hunt

Stag hunt in a normal form game notation is:

$$N = \{1, 2\}, \quad A = \{S, H\} \times \{S, H\}, \quad u(S, S) = (1, 1), \quad u(S, H) = (0, \frac{2}{3}), \quad u(H, S) = (\frac{2}{3}, 0), \quad u(H, H) = (\frac{2}{3}, \frac{2}{3})$$

Let s_1 be player 1 strategy

$$s_1 \sim \begin{pmatrix} S & H \\ p & 1-p \end{pmatrix}$$

and s_2 player 2 strategy

$$s_2 \sim \begin{pmatrix} S & H \\ q & 1-q \end{pmatrix}$$

Then it is possible to write the expected return of player 1 and 2 given the strategy profile $s = (s_1, s_2)$ by definition 2.3 :

$$e_1(s) = \sum_{a \in \{(S,S), (S,H), (H,S), (H,H)\}} u_1(a) \prod_{j=1}^n s_j(a_j) = 1(pq) + 0(p(1-q)) + \frac{2}{3}((1-p)q) + \frac{2}{3}((1-p)(1-q)) = pq - \frac{2}{3}p + \frac{2}{3}$$

Similarly for e_2 :

$$e_2(s) = pq - \frac{2}{3}q + \frac{2}{3}$$

In order to find a Pareto optimal strategy profile we can solve two constrained optimization problems where we maximize the functions $e_1(s)$ and $e_2(s)$ with constraints $p, q \in [0, 1]$.

2.3 Game 2 - Subsidy game

2.4 Game 3 - Prisoner's dilemma

2.5 Game 4 - Biased Rock-Paper-Scissors

References

- [YS09] Kevin Leyton-Brown Yoav Shoham. Multiagent systems: Algorithmic, game-theoretic, and logical foundations. 2009.